



**Hochschule
Bonn-Rhein-Sieg**
University of Applied Sciences

Master's Degree

This is to certify that
Mr Ramit Sharma,
born 15.06.1991
in Dhanusha , Nepal,

passed the
Master's Examination
on the following course:

on 11.01.2022

In acknowledgement
of this achievement,
the Bonn-Rhein-Sieg
University of Applied Sciences
hereby confers upon him the
following academic title:

Master of Science in Autonomous Systems

Master of Science

Sankt Augustin,
11.01.2022

The Dean

Prof. Dr. Wolfgang Heiden



This Diploma Supplement model was developed by the European Commission, Council of Europe and UNESCO/CEPES (United Nations Educational, Scientific and Cultural Organization /the European Centre for Higher Education Centre Européen pour l'Enseignement Supérieur). The purpose of the supplement is to provide sufficient independent data to improve the international transparency and fair academic and professional recognition of qualifications (diplomas, degrees, certificates, etc.). It is designed to provide a description of the nature, level, context, content, and status of the studies that were pursued and successfully completed by the individual named on the original qualification to which this supplement is appended. It should be free from any value judgements, equivalence statements, or suggestions about recognition. Information in all eight sections should be provided. Where information is not provided, an explanation should give the reason why.

1 Holder of the Qualification

1.1 Family Name	Sharma
1.2 First Name	Ramit
1.3 Date, Place of Birth	15 June 1991, Dhanusha , Nepal
1.4 Student ID Number	9030410

2 Qualification

2.1 Name of Qualification (full, abbreviated; in original language)	Master of Science - M. Sc.
2.2 Main Field(s) of Study	Autonomous Systems
2.3 Institution Awarding the Qualification (in original language) Status (Type / Control)	Hochschule Bonn-Rhein-Sieg Fachbereich Informatik University of Applied Sciences / State Institution Department of Computer Science
2.4 Language(s) of Instruction/Examination	English

3 Level of the Qualification

3.1 Level	Graduate/ second degree, specialization subject, and thesis
3.2 Official Length of Program	2 years
3.3 Access Requirements	Qualified Bachelor degree (three or four years), in Computer Science or appropriate related fields, or foreign equivalent (e.g. Diploma)



4 Contents and Results Gained

4.1 Mode of Study

Full-time, 2 years

4.2 Programme learning outcomes

The Master of Autonomous Systems program combines training based on theoretical foundations, broad coverage of scientific methods, and application-oriented engineering. The Master program is research-oriented and consecutive to a first degree (bachelor or equivalent) in Computer Science. The attainment of a high level of competence in the application of computer science to the field of autonomous systems is at the focus of this program.

The aim of the Master's program is to deepen and broaden the knowledge and skills of students who have completed a three year course in computer science or related. The graduates are taught scientific methods and are capable to solve complex, multi-faceted and multi-layered problems in practice as well as in the field of applied research. An integral part of the qualification in autonomous systems includes an intense study of autonomous robots and software technology in robotics and embedded systems.

The course aims to qualify graduates in the following key skills:

A high level of competence in core technical subjects of autonomous systems:

- The ability to develop qualitatively high-value, complex system with focus on it's software.
- Knowledge of developments in computer software which are on the threshold of practical application.

Methodological and scientific work:

- The ability for abstraction and model building.
- The capacity to analyze and structure complex problem areas in preparation for subsequent system development.
- The capacity to develop new methods and techniques.
- The ability to undertake application-oriented work based on specific demands and circumstances.
- The ability to select appropriate problem-specific computer science methods from a broad range of methods.

Social skills:

- The ability to transfer scientific knowledge to areas of business practice.
- The ability to present complex material in a clear and understandable way.
- The ability to head a large project or team.

The course has a duration of 2 years with a total of 120 ECTS points. It is comprised of core courses (20% of required ECTS points), elective courses (25%), seminars (10%), laboratory courses (7.5%), and a research project and colloquium (15%). The research-oriented Master thesis has a weighting of 20% of required ECTS points, the final colloquium a weighting of 2.5%. It is encouraged but not required to spend one semester abroad at a partner university. The following table surveys the ECTS credits required in each macro module:

Macro module	Semester				Sum
	1.	2.	3.	4.	
Core Courses	24				24
Electives		18	12		30
Seminars	6	6			12
Laboratory Courses		6	3		9
Research Project and Colloquium			15	3	18
Master's Thesis and Colloquium				27	27
Sum	30	30	30	30	120



4.3 Program Details

See "Masterzeugnis" (Transcript) for subjects offered in written and oral examinations and the topic of the thesis, including evaluations.

4.4 Grading Scheme

The grading scheme in Germany usually comprises five levels (with numerical equivalents; intermediate grades may be given): "Sehr gut" (1) = Very Good; "Gut" (2) = Good; "Befriedigend" (3) = Satisfactory; "Ausreichend" (4) = Sufficient; "Nicht ausreichend" (5) = Non-Sufficient/Fail. The minimum passing grade is "Ausreichend" (4). Verbal designations of grades may vary in some cases and for doctoral degrees. In addition institutions may already use the ECTS grading scheme, which operates with the levels A (best 10 %), B (next 25 %), C (next 30 %), D (next 25 %).

4.5 Overall Classification

satisfactory (2.7)

Based on written and oral examinations (75%), thesis (20%) and final colloquium (5%), cf. Masterzeugnis (Final Examination Certificate).

5 Function of the Qualification

5.1 Access to Further Study

Qualifies to apply for admission to PhD programmes.

5.2 Professional Status

The master degree in a computer discipline entitles its holder to exercise advanced professional work in the field(s) of computer science and related fields.



Certificate of Results

for the Master's Examination

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on 11.01.2022

Title of Course

Autonomous Systems

Modules

Core Courses

Advanced Software Technology
Autonomous Mobile Robots
Mathematics for Robotics and Control
Artificial Intelligence for Robotics

Grade¹

Credits

sufficient (4.0) 6
satisfactory (2.7) 6
sufficient (4.0) 6
satisfactory (3.3) 6

Specialization Intelligent Robotics

Computer Vision
Model-Driven Robot Systems and Software Engineering
Neural Networks
Learning and Adaptivity
Natural Language Processing

satisfactory (3.3) 6
satisfactory (3.0) 6
good (2.3) 6
satisfactory (2.7) 6
good (2.3) 6

Research & Development Project

Model-Driven Robot Systems and Software Engineering
Natural Language Processing
Recognizing textual entailment: Analysis of deep learning models
Research and Development Colloquium

satisfactory (3.0) 6
good (2.3) 6
good (2.3) 15
good (2.3) 3

Seminar

Introduction to Scientific Work
Advanced Scientific Working

passed 6
passed 6

Laboratory

Software Development Project
Scientific Experimentation and Evaluation

passed 6
passed 3

¹See the „Grades“ section for an explanation of German grading system.

Certificate of Results
for the Master's Examination

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of Ramit Sharma

Title of Object Detection in Dense Volume Data
Master Thesis

Thesis Prof. Dr. Paul G. Plöger
Examiners Prof. Dr. Ernst Kruijff

Grade awarded for Master's thesis	good (2.5)	24 Credits
Grade awarded for Master's colloquium	good (2.5)	3 Credits
Overall grade	satisfactory (2.7)	

Sankt Augustin,
den 11.01.2022

Chairman of the
Board of Examiners


Prof. Dr. Paul G. Plöger

The examination was conducted in accordance with the Examination Regulations of Bonn-Rhein-Sieg University of Applied Sciences dated 16 July 2009 in the latest version of 25 October 2012. The Examination Regulations were promulgated pursuant to Section 2, Subsection 4 and Section 94 of the Act Governing Universities in the Federal State of North Rhine-Westphalia (Gesetz über die Hochschulen des Landes Nordrhein-Westfalen) of 31 October 2006.

Grades 1= Excellent, 2=Good, 3=Satisfactory, 4= Sufficient
1.0 to 1.2 = „A“ Excellent
1.3 to 1.5 = „B“ Very Good
1.6 to 2.5 = „C“ Good
2.6 to 3.5 = „D“ Satisfactory
3.6 to 4.0 = „E“ Sufficient

Computation of The overall grade is calculated on the basis of a weighted average of the grades for the
overall grade examinations, the grade for the Master's thesis and the grade for the Master's colloquium*.

* Oral examination in which the candidate defends his thesis.



6 Additional Information

6.1 Additional Information

n.a.

6.2 Further Informations Status

About the institution:

<http://www.h-brs.de>

About the department:

<https://www.h-brs.de/en/inf>

About the study course:

<https://www.h-brs.de/de/inf/autonomous-systems-msc>

7 Certification

This Diploma Supplement refers to the following original documents:

Master certificate of

11 January 2022

"Zeugnis" of

11 January 2022

Transcript of

11 January 2022

Certificate Date:

11 January 2022

Dean of the Department

Prof. Dr. Wolfgang Heiden

Chairman Examination Committee

Prof. Dr. Paul G. Plöger

8 National Higher Education System

The information on the national higher education is available at:

https://www.h-brs.de/files/natstatem_ger.pdf

https://www.h-brs.de/files/natstatem_eng.pdf